

Data Science Project Report

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Abstract

Knowledge Discovery and Data Mining Tools Competition (KDD Cup'99) has been used on prior choice by researchers for the purpose of research and simulation for network intrusion detection system as well as intrusion prevention system in network security. A statistical analysis on the KDD Cup'99 dataset has been done for the objective of better understanding of structural and organizational view of the features/attributes of network traffic data. This analysis is required for evaluating various predictions on the attributes of the network traffic by taking consideration of different statistical aspects of KDD Cup'99 dataset. Code

1. Introduction

With the enormous growth of computer networks usage and the huge increase in the number of applications running on top of it, network security is becoming increasingly more important. All the computer systems suffer from security vulnerabilities which are both technically difficult and economically costly to be solved by the manufacturers. Therefore, the role of Intrusion Detection Systems (IDSs), as special-purpose devices to detect anomalies and attacks in the network, is becoming more important. The research in the intrusion detection field has been mostly focused on anomaly-based and misuse-based detection techniques for a long time. While misuse-based detection is generally favored in commercial products due to its predictability and high accuracy, in academic research anomaly detection is typically conceived as a more powerful method due to its theoretical potential for addressing novel attacks

Dataset KDD Cup'99 being a reliable IDS benchmark is a labeled intrusion detection dataset. University of California, Irvine (UCI) provides a publicly free dataset named KDD Cup'99, International Knowledge Discovery and Data Mining Tools Competition, for researchers and designers of IDS who use KDD Cup'99 as benchmark on which different methodologies are evaluated. The overall aim of this Project to classify between benign and anomalous traffic, simulating a range of cyber threats such as denial-of-service attacks, probing, remote-to-local, and user-to-root attacks to

reduce cost of security breaches. Besides this, the aims of statistical analysis are

- To understand the structure and pattern of the KDD Cup'99 dataset
- To conduct hypothesis testing for analyzing patterns and relationships within the dataset.
- To compare the classification performance of different Scaling techniques.

2. Dataset Description

The 1998 DARPA Intrusion Detection Evaluation Program, managed by MIT Lincoln Laboratory, aimed to evaluate research in network intrusion detection by providing a standard set of audit data. The dataset simulated a military network environment, including various types of intrusions, and served as the foundation for the 1999 KDD intrusion detection contest, which utilized a processed version of this data.

To create the dataset, Lincoln Laboratory simulated a typical U.S. Air Force LAN and collected nine weeks of raw TCP dump data. The LAN was operated as a realistic Air Force environment but was intentionally infused with multiple attack scenarios. A connection, defined as a sequence of TCP packets between a source and target IP address under a specific protocol, was labeled as either normal or attack, with each record approximately 100 bytes in size.

The attacks are categorized into four main types:

- **Denial-of-Service (DOS):** Attacks intended to overwhelm a system, such as *SYN flood*.
- **Remote-to-Local (R2L):** Unauthorized access from a remote machine, such as *password guessing*.
- **User-to-Root (U2R):** Unauthorized access to local superuser privileges, such as *buffer overflow* attacks.
- **Probing:** Surveillance and other scanning activities, such as *port scanning*.

In total, the dataset contains **41 features** and **4,898,431** traffic data records, of which approximately **3,925,650** are attack records.

3. Dataset Challenges and Their Solutions

- **Outliers:** Outliers were identified using the z -score method ($|z| > 3$) for the top 10 most important features, determined using a Random Forest model. Records containing such outliers were subsequently removed.
- **Class Imbalance:** There was an imbalance in the original dataset with respect to the number of samples per label. We balanced this out by undersampling from the attack class having higher number of samples to have a balanced dataset, with 972780 samples in each class.
- **Scalability:** Processing a large dataset posed computational challenges. Dimensionality reduction methods, such as SVD and JL matrices, were applied to enhance scalability.
- **Further Pre-Processing:** To prepare the data for model training, we standardized the continuous features to ensure compatibility with ML models. We also performed label encoding to transform categorical features into numeric values.

4. Hypothesis Tests and Their Conclusions

The goal of hypothesis testing in this analysis is to:

- Compare means of numerical features between normal traffic and attack traffic using Welch's T-Test.
- Examine associations between categorical features and attack types using the Chi-Square test.

We aim to infer whether certain features significantly differ between normal and attack traffic or whether they are associated with attack types. This helps in identifying the key features for classification or anomaly detection. To analyze feature significance, several hypothesis tests were conducted. We took 5000 samples, sampled at random, and performed the following tests:

4.1. Welch's T-Test

Welch's T-Test is a variation of the Student's T-Test that does not assume equal variances between groups. Since our data comes from potentially two distinct groups (normal and attack traffic), we used Welch's T-Test to compare the means of numerical features.

4.1.1 Hypotheses

- **Null Hypothesis (H_0):** There is no significant difference in the mean values of the feature between normal and attack traffic.
- **Alternative Hypothesis (H_1):** There is a significant difference in the mean values of the feature between normal and attack traffic.

4.1.2 Test Characteristics

- **Test Type:** Two-tailed test (as we are testing for any difference, not a specific direction of the difference).
- **Degrees of Freedom (df):** Calculated using the Welch-Satterthwaite formula because variances are not assumed equal.
- **Significance Level (α):** 0.05 (5% confidence level).

4.1.3 Features Tested

The following numerical features were tested: *duration*, *src_bytes*, *dst_bytes*, *count*, *srv_count*.

4.1.4 Test Results

Feature	Statistic (Sampled Data)	p-value (Sampled Data)	Significance (Sampled Data)	Chi-Square Statistic (Full Dataset)	p-value (Full Dataset)	Significance (Full Dataset)
Protocol Type (TCP)	72.87	1.38×10^{-17}	Significant	2281.20	0.0	Significant
Protocol Type (UDP)	71.87	2.30×10^{-17}	Significant	2214.37	0.0	Significant
Service (FTP)	1.43	0.23	Not Significant	25.50	4.43×10^{-7}	Significant
Service (HTTP)	2945.80	0.0	Significant	73932.58	0.0	Significant
Flag Type (SF)	4152.08	0.0	Significant	106545.26	0.0	Significant

Figure 1: T Test

4.1.5 Inference

- **Duration, dst_bytes, count:** Significant differences in mean values between normal and attack traffic, likely strong indicators for attack detection.
- **src_bytes:** Not significant in the sampled data but significant in the full dataset, indicating its importance when considering larger data.
- **srv_count:** Not significant in sampled data but significant in the full dataset, suggesting its role in attack detection.

4.2. Chi-Square Test

The Chi-Square test is used to determine whether there is a significant association between categorical variables. In this analysis, we test if certain categorical features (e.g., protocol type, service, flag) are associated with attack types.

4.2.1 Hypotheses

- **Null Hypothesis (H_0):** The feature is independent of attack type.
- **Alternative Hypothesis (H_1):** The feature is associated with attack type.

4.2.2 Test Characteristics

- **Test Type:** Two-tailed test.
- **Degrees of Freedom (df):** Determined based on the contingency table for each feature.
- **Significance Level (α):** 0.05.

4.2.3 Features Tested

The following categorical features were tested: *protocol_type_tcp*, *protocol_type_udp*, *service_ftp*, *service_http*, *flag_SF*.

4.2.4 Test Results

Feature	T-Statistic (Sampled Data)	p-value (Sampled Data)	Significance (Sampled Data)	T-Statistic (Full Dataset)	p-value (Full Dataset)	Significance (Full Dataset)
Duration	2.30	0.0215	Significant	11.76	6.72×10^{-32}	Significant
src_bytes	-0.57	0.5705	Not Significant	2.48	0.0132	Significant
dst_bytes	23.39	2.11×10^{-111}	Significant	110.78	0.0	Significant
count	-96.99	0.0	Significant	-507.54	0.0	Significant
srv_count	1.68	0.0933	Not Significant	7.23	4.78×10^{-13}	Significant

Figure 2: Chi-Square Test.

4.2.5 Inference

- **Protocol Types (TCP and UDP):** Strongly associated with attack types in both sampled and full datasets, indicating key features for identifying attacks.
- **Service (HTTP):** Highly significant association with attack type, showing that HTTP-related traffic may be prone to attacks.
- **Flag Type (SF):** Significant association with attack type, making it a crucial categorical feature.
- **Service (FTP):** Not significant in sampled data but significant in the full dataset, suggesting a minor role in attack detection.

4.3. Hypothesis Testing Results

- **Numerical Features:** Features like *duration*, *dst_bytes*, *count* have significant differences between normal and attack traffic, making them critical indicators.
- **Categorical Features:** Features like *protocol_type*, *service_http*, *flag_SF* show strong associations with attack type, highlighting their importance.

5. ML Models Used

We used the following machine learning models and trained them on our dataset, after hyper parameter tuning with the help of the Grid Search:

- **Logistic Regression:** It is a simple and interpretable model that works well for linearly separable data. It is easy to implement, fast to train, and provides probabilistic outputs, making it a good baseline model for comparison. However, its main limitation is that it may not capture complex relationships in the data, which could be crucial for detecting subtle attack patterns.
- **XG Boosting:** It is ensemble methods that build trees sequentially to correct the errors of previous trees. These models are highly effective at capturing complex patterns in the data and tend to perform very well on structured datasets like KDD'99. They can handle class imbalance and missing data gracefully, but they are computationally expensive and less interpretable than individual decision trees.

6. Scaling Techniques and Benefits

Dimensionality reduction methods played a crucial role in optimizing performance:

- **JL Matrix:** We transform the data into a lower dimension using a JL matrix, whose entries are sampled from $\mathcal{N}(0, \frac{1}{k})$, where k is the size of the reduced number of dimensions (in our case, $k = 10$). The JL lemma ensures that the distance between points is approximately preserved even after transformation, which helps maintain the structure of the data. We also experiment with a sparse JL matrix, where the default density is $\frac{1}{\sqrt{n}}$, to reduce computation time.
- **PCA using SVD:** We perform Principal Component Analysis (PCA) by applying Singular Value Decomposition (SVD) to the data. The data is projected onto the first k principal components, where $k = 10$ in our case. This dimensionality reduction technique captures the most significant variance in the data while reducing its dimensionality. SVD is computed efficiently, and the

data is transformed to the reduced subspace, preserving its important structural features.

- **Coresets for KMeans:** Coresets are small, weighted summaries of large datasets such that solutions found on the summary are provably competitive with solutions found on the full dataset for specific ML tasks, KMeans Clustering in our case. In our approach, we first experiment with various coreset sizes and, after iterative testing, determine that the optimal coreset size is 5,000, which corresponds to 1% of the original dataset size. This coreset was created using D2 sampling, ensuring that the selected subset retains the key information necessary for the task at hand.

7. Model Performance on Scaled and Unscaled Datasets

7.1. Unscaled Dataset

Algorithm	Train Accuracy	Test Accuracy	Runtime
Logistic Regression	99.66	99.63	3.4
XGBoost	100	99.99	36.2

Table 1: Performance on Unscaled Dataset

7.2. Scaled Dataset

- **PCA:**

Algorithm	Train Accuracy	Test Accuracy	Runtime
Logistic Regression	98.52	98.43	0.6
XGBoost	99.98	99.93	18.2s

Table 2: Performance on Scaled Dataset using PCA

- **Regular JL:**

Algorithm	Train Accuracy	Test Accuracy	Runtime
Logistic Regression	98.46	98.43	7.8
XGBoost	99.97	99.91	63.0

Table 3: Performance on Scaled Dataset using Regular JL

- **Sparse JL:**

Algorithm	Train Accuracy	Test Accuracy	Runtime
Logistic Regression	97.77	97.79	6.2
XGBoost	100	99.97	52.8

Table 4: Performance on Scaled Dataset using Sparse JL

- **Coresets:**

Metric	Value
Sum of weights of X	494021.0
Sum of weights of C	489069.5046560087
Cost(X, Q^*_c)	2209250.255535879
Cost(C, Q^*_c)	2153293.030875129
Cost(X, Q^*_x)	2203843.0870299623
Cost(C, Q^*_x)	2160770.6739207683

Table 5: Summary for Coreset

8. Conclusion and Future Work

We successfully developed a Network Intrusion Detection model with high accuracy and efficiency. Key take-aways include:

- XG Boosting consistently outperformed Logistic Regression in classification tasks, but it is not lightweight
- PCA reduced runtime significantly while maintaining a very high accuracy.
- JL Scaling, didn't improve efficiency substantially. However, sparse JL was found to be better than the regular matrix when XGBoost was used (100% accuracy on training data)
- Coreset performed equally well as full dataset for KMeans clustering

Future Directions:

- Explore advanced machine learning and deep learning models, such as autoencoders, for improved scalability and accuracy.
- Incorporate real-time network analysis for dynamic intrusion detection.

By advancing detection methods and promoting user awareness, we can create a safer online environment.

References

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